PH7002: Concepts in Statistical Mechanics
§0 Administrivia

Asst. Prof. Yong Ee Hou
Nanyang Technological University
AY2025-2026 Sem 1
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§0.1 Course Aim

§0.2 Course Instructor

§0.3 Course Schedule

§0.4 Assessment and Policy
- §0.4.1 Weightage
- §0.4.2 Assignments
- §0.4.3 Online Assessment
- §0.4.4 Midterm
- §0.4.5 Final Project

§0.5 References

§0.6 Copyright

Course Aim

Welcome to **PH7002:** *Concepts in Statistical Mechanics*. This is an introductory course on the **physics of phase transitions** and acts as a first stepping stone towards an understanding of phase transitions for those beginning research. You will use knowledge from previous courses in noninteracting systems to develop the theoretical framework of interacting systems. Through the course, you will build foundational knowledge in key topics in statistical mechanics such as mean field theory, transfer matrix, Monte Carlo simulations, renormalization group, which are critical in the study of phase transitions.

Please refer to OBTL on NTU PAP course information website for more information.

Course Instructor

- **Name:** Asst. Prof. YONG Ee Hou
- **Email:** eehou@ntu.edu.sg

  Feel free to drop me an email if you have questions regarding the course.
- **Website:** https://personal.ntu.edu.sg/eehou/
- **Office:** SPMS-PAP-04-05
- **Office hours:** Thu 3:30pm–4:30pm, week 2 to week 13, excluding recess week.

Course Schedule

Class times are as follows:

Type	Day	Time	Venue
Lecture	Tuesday	10:30am–12:30pm	SPMS TR+15
Lecture	Thursday	1:30pm–2:30pm	SPMS TR+15
Tutorial	Thursday	2:30pm–3:30pm	SPMS TR+15

Because weekly class attendance and success in class are **positively correlated**, students are expected to **participate** in lectures and tutorials. Lecture participations has a weightage of 10% of the course and will be done using WOOCLAP, accessible from NTULearn.

Lecture

- Course materials will be **self contained** as much as possible.
- **Find what works for you**. Some of you will learn from lectures, notes, each other.
- **The sky is the limit**. You are encouraged to explore and learn more on your own.
- Please ask questions during class! I love it when students asked questions during class.

Weightage

Assessment		
Type	Weightage	Comments
Assignments	20%	∼Biweekly (5 in total)
†Online Assessment	10%	Every lecture
Midterm	20%	**9th Oct 2025 (Thu)**, in-class ~ Thu 1:30pm to 2:30pm
Final Project	50%	**21st Nov 2025 (Fri)**

Please add the stated in-class midterm time to your calendar and inform me of any conflicts *beforehand*. You may bring a double sided formula sheet and calculator.

† Online Assessment is conducted via **Wooclap**.

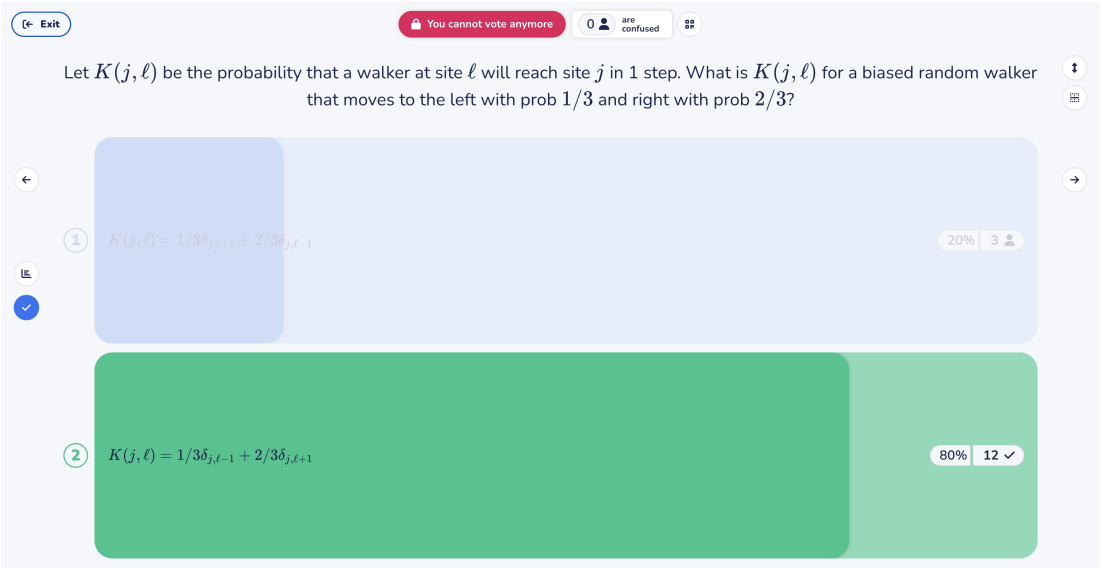
Assignments

Group discussion is encouraged when attempting the assignments (20% of total grade). However, please complete the homework yourself. Homework assignments will be distributed regularly, about once in 2 weeks, a total of 5 assignments.

Each assignment will have a due date, and you are required to upload to **NTULearn** in **ONE** PDF file **BEFORE** and upright orientation before the deadline. Please let me know in advance if you know that you will have a conflict with the due date (e.g. illness, scientific conferences, family issues, etc) to avoid any penalty.

Online Assessment

WOOCLAP in-class participation (10% of total grade). There will be questions asked using Wooclap during lectures to promote *active learning*. Please login to WOOCLAP from the link provided in NTULEARN to ensure proper registration of your **NTU credentials**. This will ensure WOOCLAP can transfer your attendance and marks back to NTULearn properly.



Midterm

- Midterm (20% of total grade) will be in class during the middle of the semester around recess week.
- It will be a **RESTRICTED OPEN BOOK** exam. You are only allowed to bring in **ONE (1)** sheet of **A4 note HAND-WRITTEN/TYPED** on both sides.
- Calculators may be used. However, you should write down systematically the steps in the workings to get full credit.
- You need an official leave authorized by the school to avoid getting 0% for any missed midterm.
- I will plan for a **Supplementary Assessment (SA)** at the end of the term for students who miss the midterm. This SA is **compulsory** to those who missed the midterm with valid leave.

Final Project

- For the final project (50% of total grade), you will be required to do a research on a topic of interest and present your findings to the class.
- You should consult me regarding the choice of topic (ideally around 3rd to 5th week) to ensure it is suitable and also it is different from your classmates.
- Duplicate project topics should be avoided.
- There will be three components to the assessment, namely, **short essay (35%)**, **interactive notebook (15%)**, and **final presentation (30% slides + 20% presentation)**. I will provide more information in a separate document.
- Please refer to the project sheet for more information.

Goal of Final Project

- Demonstrate **good taste, curiosity and ambition** in your choice of subject
- Capable of **distilling** the most important and essential details from very technical papers
- **Develop skills** in numerical methods that are increasingly important for both academia and industry
- Develop your **presentation skills** – how you deliver the content.
- Learn how to write technical papers for submission to journals such as PRL, PRE, etc.

Final Project

You need to submit to NTULearn your the final project, which includes the following files:

1. PDF output of your paper (35%).
2. Powerpoint/keynote/PDF slides (30%).
3. Turnitin report.
4. Interactive jupyter notebook or mathematica notebook (15%).

The final project is due **21st Nov 2025 (Friday)**. The presentations will take place in class during **week 13**. The talks must be attended by all students.

Turnitin

- Please submit a **Turnitin report together** with the term paper; failure to do so will result in a 15% penalty.
- The total similarity index from the turnitin report should ideally be less than 15% and strictly less than 20%. Similarly for the AI index.
- Anything above 20% will result in a penalty that works as follows: if the similarity/AI index is 25%, there will be a penalty of (25-20) = 5%; if the similarity/AI index is 30%, the penalty becomes (30-20) = 20% and so on. Penalty for exceeding similarity/AI threshold is additive.
- The use of generative AI is permitted. However, it should be used with prudence and you need to ensure your paper satisfies the AI detection limit in Turnitin. Please read Nature survey on use of AI for some current opinions.

Turnitin

N = 3.8 10⁶
N = 3.3 10⁶
N = 1.9 10⁵
FIG. 1. Non-destructive in trap images of the atomic cloud taken at different stages of the experiment with a) static BEC serving as a baseline of aspect ratio being R_x/R_y = 1.57 b) subsequent image of BEC cloud after the employment of evaporative spin-up with its aspect ratio implying $N = 3.3 \times 10^6, \tilde{\Omega} = 0.953$ and, c) with optical spin-up, further speeding up the cloud into N 1.9 × 10⁵, $\tilde{\Omega}$ = 0.993. Compared to the static image, the centrifugal force distorts the cloud into an oblate shape affecting the aspect ratio to decrease by a factor of 8.
II. METHODS & RESULTS
To rotate the BEC, the authors use a so-called "evaporative spin up" technique used previously in [6]. In this technique, they start with an order of 10⁶ Rb-87 atoms in the hyperfine state, pre-cooled to slightly higher than the critical temperature of its BEC, and placed in a time-orbiting potential (TOP) magnetic trap which is a static quadrupole field with a small rotating radial bias field[7]. The effective average potential is axially symmetric and harmonic, with axial trap frequency $\omega_z = 2\pi(5.3)$Hz and radial trap frequency $\omega_\rho = 2\pi(8.3)$Hz. Rotation of the cloud of atoms around the vertical z axis is induced by elliptically contorting the magnetic trap in the horizontal xy plane, then abruptly altering the angle of the deformation. The trap is then returned to its axial symmetry along the z axis and undergoes a quasi one-dimensional evaporation which selectively expels atoms near the z axis of rotation, thus leaving the remaining cloud with higher angular momentum, called "spinning up" the cloud. Using this technique, the authors are able to achieve a rotation rate of 95% of the radial trap frequency. The radial trap frequency is also the centrifugal rotation limit for the harmonically trapped cloud, past which the cloud will be spun out of the trap. The rotation angular frequency is denoted by Ω and is expressed as a fraction of the centrifugal rotation limit as a scaled rotation rate $\tilde{\Omega} = 0.95$ where $\tilde{\Omega} = \Omega/\omega_\rho$. At the point when $\tilde{\Omega} = 0.95$, about half of the atoms survive to produce the fast rotating BEC.
The rotation creates centrifugal forces causing the cloud to bulge radially outwards. Thus, changes in the aspect ratio of the cloud from the side view enable the calculation of the rotation rate. A static cloud has an aspect ratio 1.53 while the rotating cloud has an aspect ratio 0.48. This can be visualized in 1.
The problem with using only the "evaporative spin up"
atoms from the cloud. The angular momentum per particle remains the same, but the loss of atoms causes the cloud of BEC to shrink in radius, which as a result of the conservation of angular momentum leads to a further increase in the scaled rotation rate $\tilde{\Omega}$. Using this "optical spin up" technique for up to two seconds, the authors decrease the number of atoms, leaving an order of 10⁵ atoms while increasing $\tilde{\Omega}$ to more than 0.99, up from the 0.95 attained with only the "evaporative spin up" technique.
Importantly, the increased centrifugal force from the increased rotation pulls the cloud radially apart even flatter into a pancake shape, reducing the density of the BEC greatly, and causing it to approach the quasi two dimensional regime. This weakens the effective inter-particle interaction of the BEC cloud and the BEC behaves almost like an ideal gas. At this point, as explained in the introduction, the condensate mainly resides in the LLL single particle states implying its existence in the mean-field quantum Hall regime. It was also predicted earlier by [8] that the vortex lattice consequently undergoes a sort of phase transition characterized by its elastic shear strength. The elastic shear strength of the lattice drastically plunges in value from the Thomas-Fermi (TF) limit, at which the condensate behaves effectively as an incompressible fluid under rotation, to a value corresponding to the mean-field quantum Hall regime. Thus, the authors use the elastic shear strength as a proxy of measurement to show that the BEC displays quantum Hall-like properties at this rapid rate of rotation.
They directly observe the elastic shear strength by exciting and imaging the so-called transverse elastic "Tkachenko modes". The experimental method for this was first introduced in [9]. The authors use a detuned laser with wavelength of 850nm aimed at the middle of the BEC cloud, along the z axis, pulling the atoms in towards the center. At the same time, the Coriolis force from the rotation deflects the direct inward path of the in-moving atoms. The laser is then switched off and the azimuthal lattice oscillations are driven by the lattice's elastic shear strength. Since these oscillations can be observed directly through imaging, the oscillation frequencies are measured by changing the time after stopping the optical pump and then observing the cycle of the lattice oscillation.
Comparing the Tkachenko frequency measurements for different rotation rates to the theoretical calculations done by Baym [8], the authors found that the Tkachenko mode frequency falls from the predicted value for the TF regime at rotation rates $\tilde{\Omega} < 0.98$, to the predicted value for the mean-field quantum Hall (or LLL) regime
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References

Required textbooks

The required textbooks for the course:

1. J. Sethna, *Statistical Mechanics Entropy, Order Parameters, and Complexity*.
   Available for download at NTULearn.
2. J. M. Yeomans, *Statistical Mechanics of Phase Transitions*.
   Please purchase a copy from the NTU bookstore.

The course will cover part of (1) and most of (2). My notes should supplemented by materials from (1) and problems assigned will come from both (1) and (2). My lectures are designed to be self-sufficient. You are encouraged to take notes during class.

References

Additional references

A number of useful reference books may be found in the library. These include:

1. D. Chandler *Introduction to modern statistical mechanics.*
2. M. Plischke and B. Bergerson, *Equilibrium Statistical Physics.*
3. N. Goldenfeld, *Lectures On Phase Transitions And The Renormalization Group.*
4. M. Kardar, *Statistical Physics of Particles.*
5. M. Kardar, *Statistical Physics of Fields.*

Useful references

Also, for students who wishes to learn more on your own, there are also alot of useful resources online. Here are some that I like:

1. Lectures on statistical mechanics by Leonard Susskind
2. Lecture notes by Michael Peskin
3. Statistical Mechanics of Particles by Mehran Kardar (MIT 1)
4. Statistical Physics of Fields by Mehran Kardar (MIT 2)

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